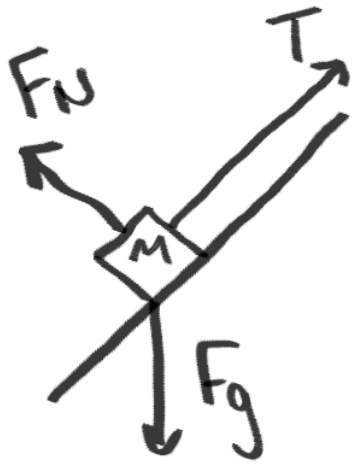
Changing frame of reference

October 3, 2026

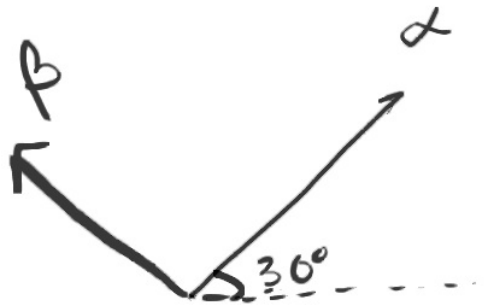
1 Change reference frame

Suppose you have a mass M on an inclined slope, 30 degrees up from the horizon. The mass is supported in place by a string with Tensile Force T.



1. Write out the two x and y equations of motion in the usual x-y coordinate system, knowing that the block is stationary.
2. If only the mass were specified, say $M = 50kg$, would you be able to use just one equation to solve for either the tensile force or the normal force?

3. Consider this new coordinate system, with α replacing x and β replacing y.



$T_\alpha = T$

$F_{N_\beta} = F_N$

Note that in this coordinate system, $T_\alpha = T$ because T is perfectly in line with the alpha axis. $F_{N\beta} = F_N$ because the normal force is perfectly in line with the beta axis. Rewrite the two equations of motion for the alpha and beta axes. For instance, $\Sigma F_{alpha} = 0 = T - F_{g\alpha}$. What is the β equation?

4. If only the mass were specified as $M = 50kg$ and you didn't know the tensile force or the normal force, solve for them.

2 BONUS

In either coordinate system we have two equations and two unknowns, which means we have enough information to completely solve it, but it wasn't as easy in the x-y coordinate system. Can you find an expression for F_N in the x-y coordinate system?